Design of Parallel Helical Gears

- Similar to spur gears, but with teeth making a *helix angle* with respect to the gear centerline
- Adds axial force component to shaft and bearings
- Smoother transition of force between mating teeth due to gradual engagement and disengagement

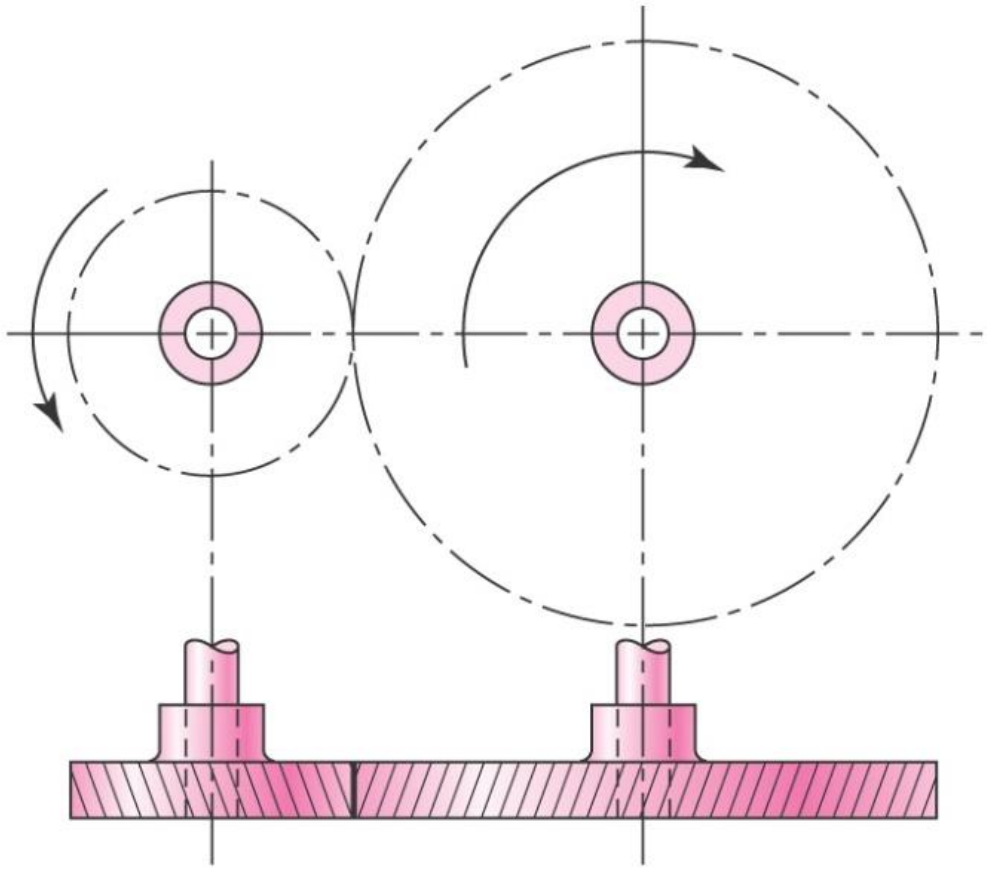


Fig. 13–2

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Parallel Helical Gears

- Tooth shape is involute helicoid

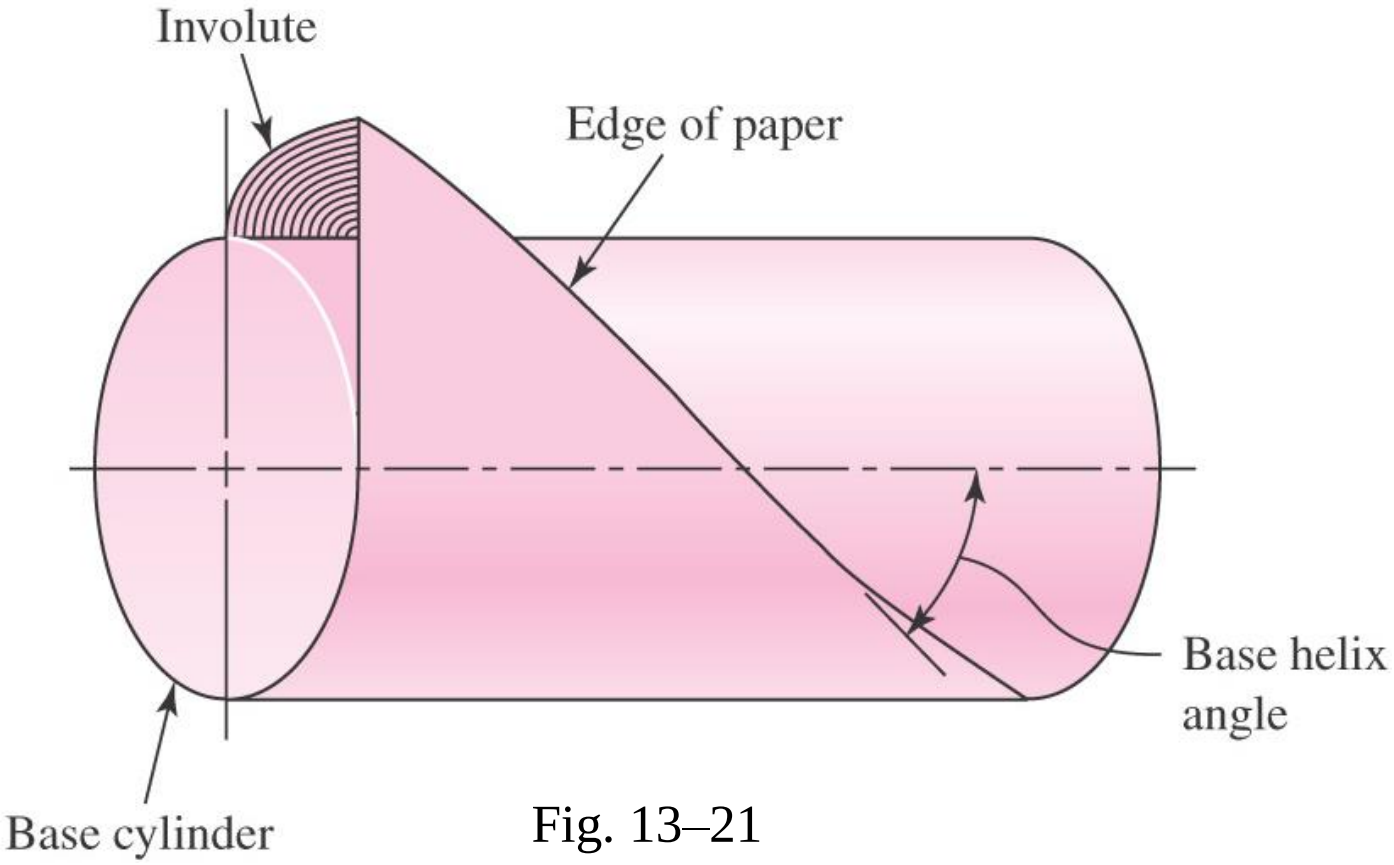


Fig. 13–21

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Force Analysis – Helical Gearing

$$W_r = W \sin \phi_n$$

$$W_t = W \cos \phi_n \cos \psi \qquad (13\text{–}39)$$

$$W_a = W \cos \phi_n \sin \psi$$

$$W_r = W_t \tan \phi_t$$

$$W_a = W_t \tan \psi \qquad (13\text{–}40)$$

$$W = \frac{W_t}{\cos \phi_n \cos \psi}$$

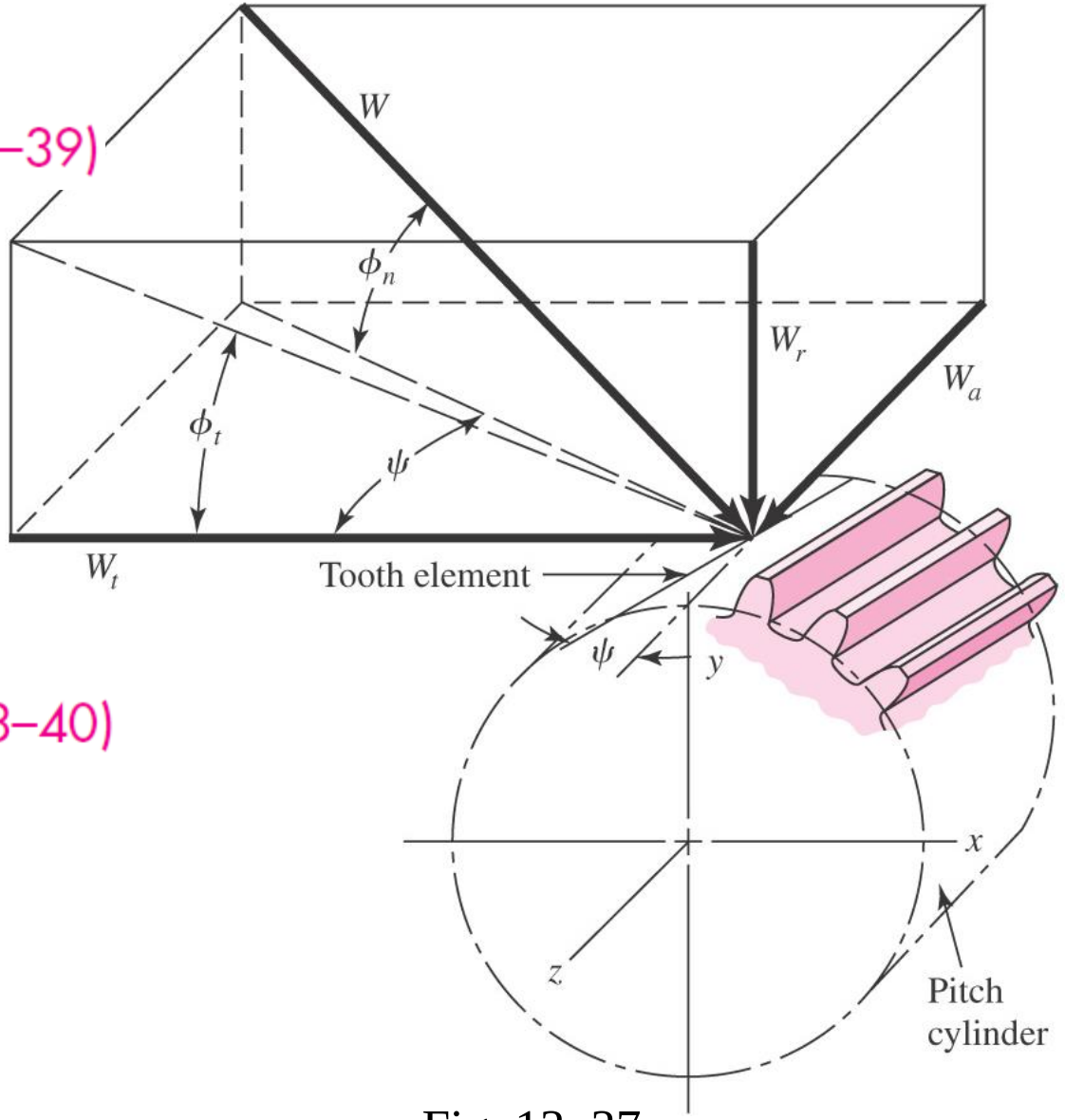


Fig. 13–37

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Parallel Helical Gears

- *Transverse circular pitch* p_t is in the plane of rotation
- *Normal circular pitch* p_n is in the plane perpendicular to the teeth

$$p_n = p_t \cos \psi \qquad (13\text{–}16)$$

- *Axial pitch* p_x is along the direction of the shaft axis

$$p_x = \frac{p_t}{\tan \psi} \qquad (13\text{–}17)$$

- *Normal diametral pitch*

$$P_n = \frac{P_t}{\cos \psi} \qquad (13\text{–}18)$$

$$p_n P_n = \pi$$

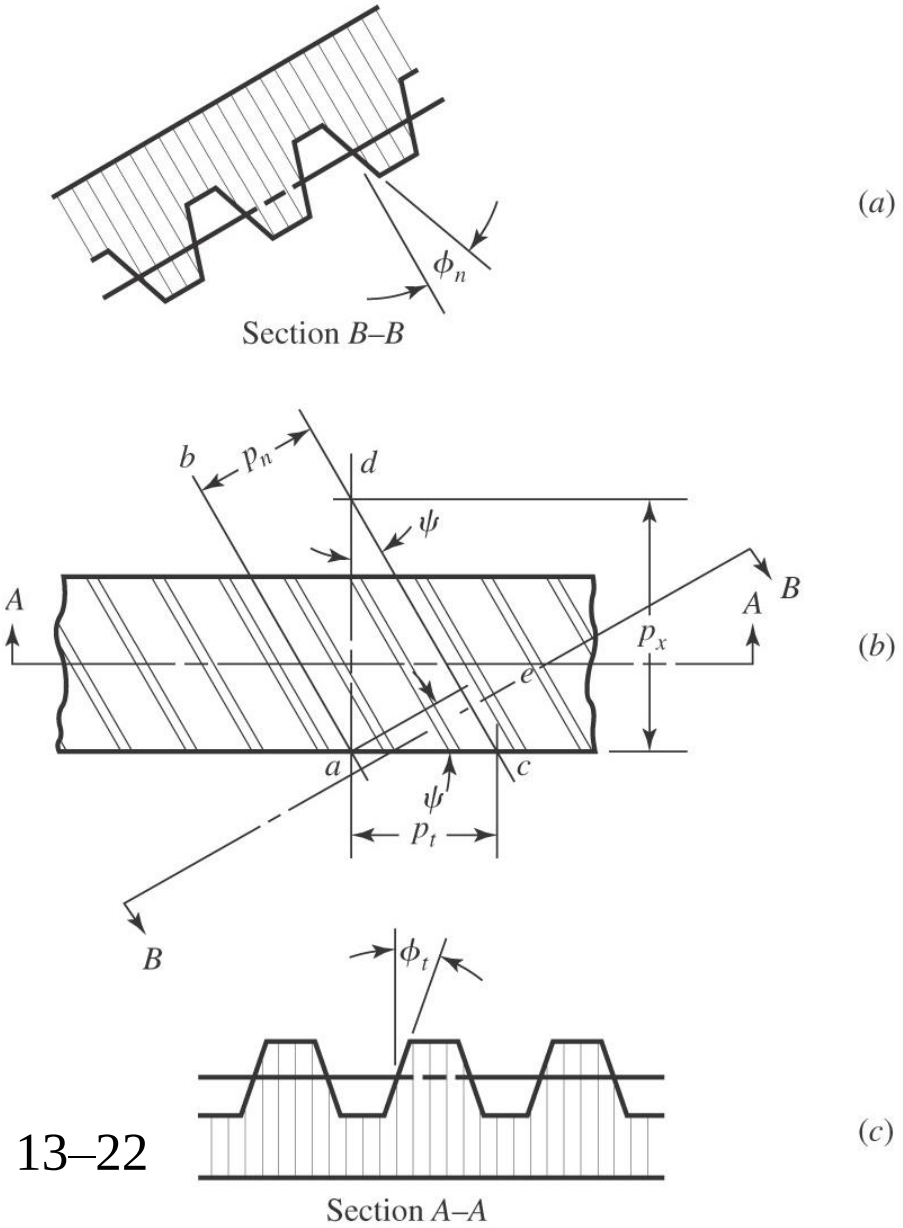


Fig. 13–22

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Parallel Helical Gears

- Relationship between angles

$$\cos\psi = \frac{\tan\phi_n}{\tan\phi_t} \qquad (13\text{–}19)$$

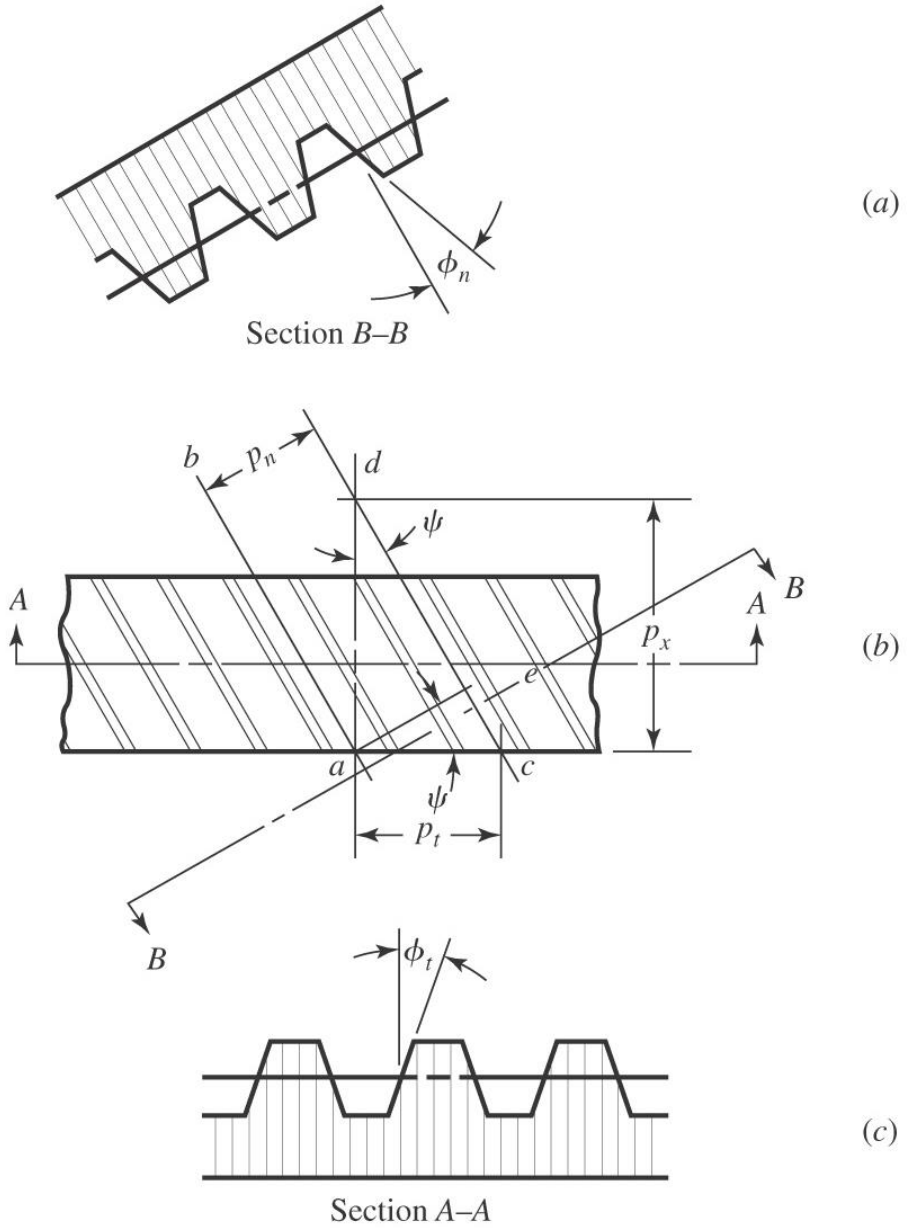


Fig. 13–22

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Example 6 - Helical Gear Geometric Parameters

A stock helical gear has a normal pressure angle of 20°, a helix angle of 25°, and a transverse diametral pitch of 6 teeth/in, and has 18 teeth. Find:
(*a*) The pitch diameter
(*b*) The transverse, the normal, and the axial pitches
(*c*) The normal diametral pitch
(*d*) The transverse pressure angle

a. $d = N/P_t$

b. $p_t P_t = \pi$ $\qquad p_n = p_t \cos\psi \qquad p_x = \dfrac{p_t}{\tan\psi}$

c. $P_n = \dfrac{P_t}{\cos\psi}$ $\qquad$ OR $\qquad p_n P_n = \pi$

d. $\cos\psi = \dfrac{\tan\phi_n}{\tan\phi_t}$

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Example 6 - Helical Gear Geometric Parameters

Solution

(a)
$$d = \frac{N}{P_t} = \frac{18}{6} = 3 \text{ in}$$
Answer

(b)
$$p_t = \frac{\pi}{P_t} = \frac{\pi}{6} = 0.5236 \text{ in}$$
Answer

$$p_n = p_t \cos\psi = 0.5236 \cos 25^\circ = 0.4745 \text{ in}$$
Answer

$$p_x = \frac{p_t}{\tan\psi} = \frac{0.5236}{\tan 45^\circ} = 1.123 \text{ in}$$
Answer

(c)
$$P_n = \frac{P_t}{\cos\psi} = \frac{6}{\cos 25^\circ} = 6.620 \text{ teeth/in}$$
Answer

(d)
$$\phi_t = \tan^{-1}\left(\frac{\tan\phi_n}{\cos\psi}\right) = \tan^{-1}\left(\frac{\tan 20^\circ}{\cos 25^\circ}\right) = 21.88^\circ$$
Answer

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AGMA Bending Stress

$$\sigma = \begin{cases} W^t K_o K_v K_s \dfrac{P_d}{F} \dfrac{K_m K_B}{J} & \text{(U.S. customary units)} \\ W^t K_o K_v K_s \dfrac{1}{b m_t} \dfrac{K_H K_B}{Y_J} & \text{(SI units)} \end{cases} \tag{14–15}$$

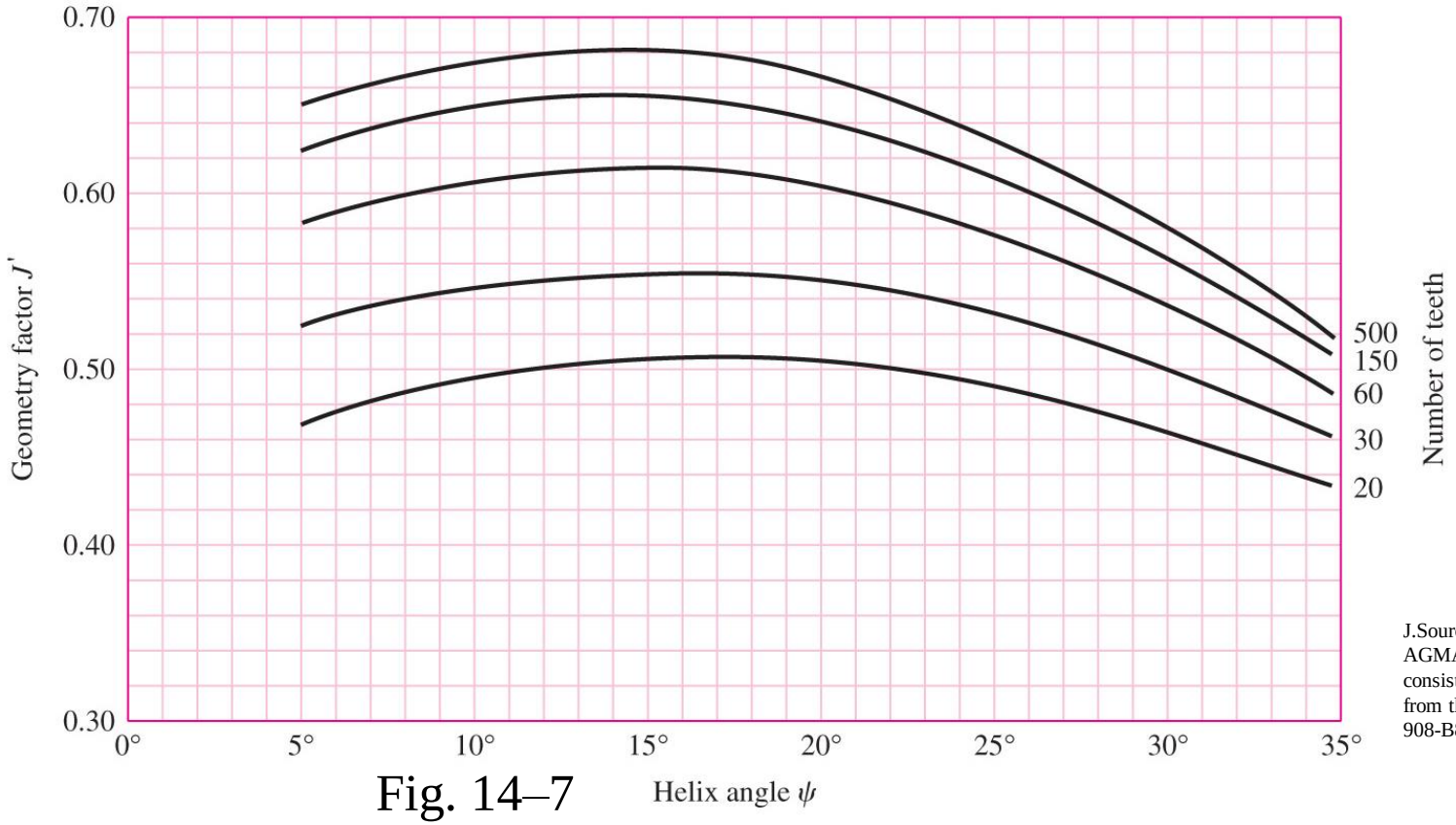
where for U.S. customary units (SI units),

W^t is the tangential transmitted load, lbf (N)
K_o is the overload factor
K_v is the dynamic factor
K_s is the size factor
P_d is the transverse diametral pitch
F (b) is the face width of the narrower member, in (mm)
K_m (K_H) is the load-distribution factor
K_B is the rim-thickness factor
J (Y_J) is the geometry factor for bending strength (which includes root fillet stress-concentration factor K_f)
(m_t) is the transverse metric module

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Helical-Gear Geometry Factor *J*

- Get *J′* from Fig. 14–7, which assumes the mating gear has 75 teeth
- Get multiplier from Fig. 14–8 for mating gear with other than 75 teeth
- Obtain *J* by applying multiplier to *J′*

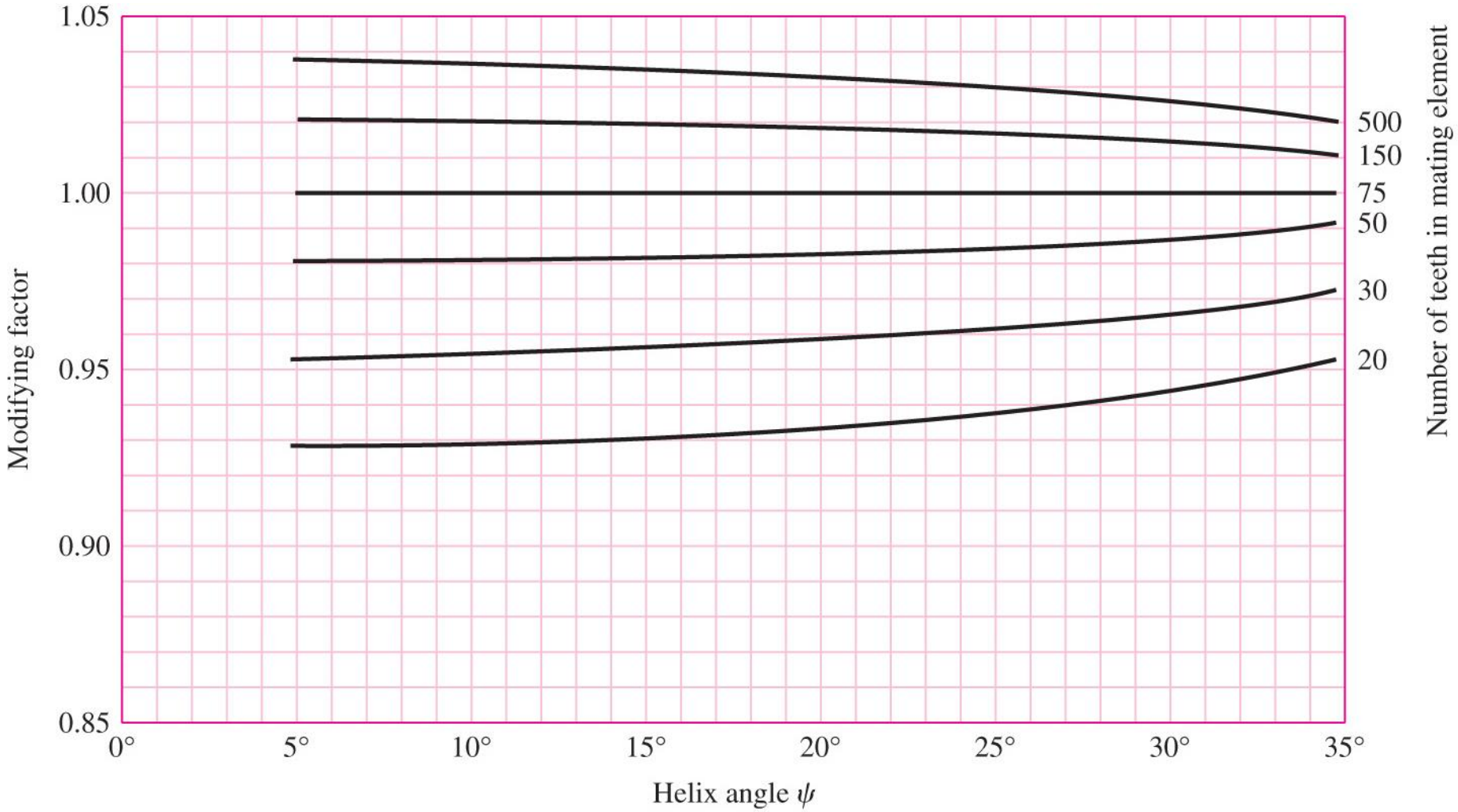


J.Source: The graph is from AGMA 218.01, which is consistent with tabular data from the current AGMA 908-B89.

Fig. 14–7

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Modifying Factor for *J*



J.Source: The graph is from AGMA 218.01, which is consistent with tabular data from the current AGMA 908-B89.

Fig. 14–8

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AGMA Contact Stress

$$\sigma_c = \begin{cases} C_p \sqrt{W^t K_o K_v K_s \dfrac{K_m}{d_P F} \dfrac{C_f}{I}} & \text{(U.S. customary units)} \\ Z_E \sqrt{W^t K_o K_v K_s \dfrac{K_H}{d_{w1} b} \dfrac{Z_R}{Z_I}} & \text{(SI units)} \end{cases} \tag{14–16}$$

where W^t, K_o, K_v, K_s, K_m, F, and b are the same terms as defined for Eq. (14–15). For U.S. customary units (SI units), the additional terms are

C_p (Z_E) is an elastic coefficient, $\sqrt{\text{lbf/in}^2}$ ($\sqrt{\text{N/mm}^2}$)
C_f (Z_R) is the surface condition factor
d_P (d_{w1}) is the pitch diameter of the *pinion,* in (mm)
I (Z_I) is the geometry factor for pitting resistance

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Surface Strength Geometry Factor I (Z_I in metric)

- Called *pitting resistance geometry factor* by AGMA

$$I = \begin{cases} \dfrac{\cos\phi_t \sin\phi_t}{2m_N} \dfrac{m_G}{m_G + 1} & \text{external gears} \\ \dfrac{\cos\phi_t \sin\phi_t}{2m_N} \dfrac{m_G}{m_G - 1} & \text{internal gears} \end{cases} \tag{14–23}$$

$$m_G = \frac{N_G}{N_P} = \frac{d_G}{d_P} \tag{14–22}$$

$$m_N = \frac{p_N}{0.95Z} \tag{14–21}$$

$$p_N = p_n \cos\phi_n \tag{14–24}$$

$$Z = \left[(r_P + a)^2 - r_{bP}^2\right]^{1/2} + \left[(r_G + a)^2 - r_{bG}^2\right]^{1/2} - (r_P + r_G)\sin\phi_t \tag{14–25}$$

$$r_b = r\cos\phi_t \tag{14–26}$$

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Allowable Bending Stress

$$\sigma_{\text{all}} = \begin{cases} \dfrac{S_t}{S_F}\dfrac{Y_N}{K_T K_R} & \text{(U.S. customary units)} \\ \dfrac{S_t}{S_F}\dfrac{Y_N}{Y_\theta Y_Z} & \text{(SI units)} \end{cases} \tag{14–17}$$

where for U.S. customary units (SI units),

S_t is the allowable bending stress, lbf/in^2 (N/mm^2)
Y_N is the stress-cycle factor for bending stress
K_T (Y_θ) are the temperature factors
K_R (Y_Z) are the reliability factors
S_F is the AGMA factor of safety, a stress ratio

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Allowable Contact Stress

$$\sigma_{c,\text{all}} = \begin{cases} \dfrac{S_c}{S_H}\dfrac{Z_N C_H}{K_T K_R} & \text{(U.S. customary units)} \\ \dfrac{S_c}{S_H}\dfrac{Z_N Z_W}{Y_\theta Y_Z} & \text{(SI units)} \end{cases} \tag{14–18}$$

S_c is the allowable contact stress, lbf/in^2 (N/mm^2)
Z_N is the stress-cycle factor
C_H (Z_W) are the hardness ratio factors for pitting resistance
K_T (Y_θ) are the temperature factors
K_R (Y_Z) are the reliability factors
S_H is the AGMA factor of safety, a stress ratio

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Example 7 - Helical Gear Tooth Design

A 17-tooth 20° normal pitch-angle helical pinion with a right-hand helix angle of 30° rotates at 1800 rev/min when transmitting 4 hp to a 52-tooth helical gear. The normal diametral pitch is 10 teeth/in, the face width is 1.5 in, and the set has a quality number of 6. The gears are straddle-mounted with bearings immediately adjacent. The pinion and gear are made from a through-hardened steel with surface and core hardnesses of 240 Brinell on the pinion and surface and core hardnesses of 200 Brinell on the gear. The transmission is smooth, connecting an electric motor and a centrifugal pump. Assume a pinion life of 10^8 cycles and a reliability of 0.9 and use the upper curves in Figs. 14–14 and 14–15.

(*a*) Find the factors of safety of the gears in bending.

(*b*) Find the factors of safety of the gears in wear.

(*c*) By examining the factors of safety identify the threat to each gear and to the mesh.

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